PRACTICAL NUMBER: 8(ii)

Aim: Implement an Algorithm for detecting cycle in given graph G

ii. Accept graph as input in the form of adjacency list

Program: -

```
//SHEIKH WASIMUDDIN 60 23071044 CSEIOT//

#include <stdio.h>
#include <stdbool.h>

#define MAX 100

int adj[MAX][MAX];
int visited[MAX];
int n;


void createGraph() {
int i, j, edges, origin, dest;

printf("Enter the number of vertices:");
scanf("%d", &n);

for (i = 0; i<n; i++) {
for (j = 0; j<n; j++) {
adj[i][j] = 0;
}
}

printf("Enter the number of edges: ");
scanf("%d", &edges);
```

```
for (i = 0; i<edges; i++) {
printf("Enter edge (origin destination):");
scanf("%d %d", &origin, &dest);

if (origin >= n || dest >= n || origin < 0 || dest < 0) {
printf("Invalid edge!\n");
i--;
} else {
adj[origin][dest] = 1;
adj[dest][origin] = 1;
}
}
}

bool isCyclicUtil(int v, int parent) {
visited[v] = 1;

for (int i = 0; i<n; i++) {
if (adj[v][i]) {
if (!visited[i]==1) {
if (isCyclicUtil(i, v))
return true;
} else if (i != parent) {
return true;
}
}
}
return false;
}

bool isCyclic() {


for (int i = 0; i<n; i++) {
visited[i] = 0;
```

```
    }


    for (int i = 0; i<n; i++) {
    if (!visited[i]==1) {
    if (isCyclicUtil(i, -1)==1)
    return true;
    }
    }

    return false;
    }

    int main() {
    createGraph();

    if (isCyclic()==1) {
    printf("The graph contains a cycle.\n");
    } else {
    printf("The graph does not contain a cycle.\n");
    }

    return 0;
}
```

Output: -

main.c

```c
//SHEIKH WASIMUDDIN 60 23071044 CSEIOT//

#include <stdio.h>
#include <stdbool.h>

#define MAX 100

int adj[MAX][MAX];
int visited[MAX];
int n;


void createGraph() {
int i, j, edges, origin, dest;

printf("Enter the number of vertices:");
scanf("%d", &n);

for (i = 0; i<n; i++) {
for (j = 0; j<n; j++) {
adj[i][j] = 0;
}
}

printf("Enter the number of edges: ");
scanf("%d", &edges);

for (i = 0; i<edges; i++) {
printf("Enter edge (origin destination):");
scanf("%d %d", &origin, &dest);

if (origin >= n || dest >= n || origin < 0 || dest < 0) {
printf("Invalid edge!\n");
i--;
} else {
```

```
Enter the number of vertices:4
Enter the number of edges: 4
Enter edge (origin destination):0 1
Enter edge (origin destination):1 2
Enter edge (origin destination):2 3
Enter edge (origin destination):3 0
The graph contains a cycle.


...Program finished with exit code 0
Press ENTER to exit console.
```

Conclusion:

Hence, Implement an Algorithm for detecting cycle in given graph Gsuccessfully executed.